

● HARD

● INFORMATION AND IDEAS

Information and Ideas

12

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell's electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
- B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.
- C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.
- D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

King Lear is a circa 1606 play by William Shakespeare. In the play, the character of King Lear attempts to test his three daughters' devotion to him. He later expresses regret for his actions, as is evident when he _____ blank

Which choice most effectively uses a quotation from *King Lear* to illustrate the claim?

- A. says of himself, "I am a man / more sinned against than sinning."
- B. says during a growing storm, "This tempest will not give me leave to ponder / On things would hurt me more."
- C. says to himself while striking his head, "Beat at this gate that let thy folly in / And thy dear judgement out!"
- D. says of himself, "I will do such things— / What they are yet, I know not; but they shall be / The terrors of the earth!"

Aerogels are highly porous foams consisting mainly of tiny air pockets within a solidified gel. These lightweight materials are often applied to spacecraft and other equipment required to withstand extreme conditions, as they provide excellent insulation despite typically being brittle and eventually fracturing due to degradation from repeated exposure to high heat. Now, Xiangfeng Duan of the University of California, Los Angeles, and colleagues have developed an aerogel with uniquely flexible properties. Unlike earlier aerogels, Duan's team's material contracts rather than expands when heated and fully recovers after compressing to just 5% of its original volume, suggesting that _____ blank

Which choice most logically completes the text?

- A. the aerogel's remarkable flexibility results from its higher proportion of air pockets to solidified gel as compared to other aerogels.
- B. the aerogel's overall strength is greater than that of other insulators but its ability to withstand exposure to intense heat is lower.
- C. the aerogel will be more effective as an insulator for uses that involve gradual temperature shifts than for those that involve rapid heat increases.
- D. the aerogel will be less prone to the structural weakness that ultimately causes most other aerogels to break down with use.

In a study of new technology adoption, Davit Marikyan et al. examined negative disconfirmation (which occurs when experiences fall short of one's expectations) to determine whether it could lead to positive outcomes for users. The team focused on established users of "smart home" technology, which presents inherent utilization challenges but tends to attract users with high expectations, often leading to feelings of dissonance. The researchers found that many users employed cognitive mechanisms to mitigate those feelings, ultimately reversing their initial sense of disappointment.

Which choice best states the main idea of the text?

- A. Research suggests that most users of smart home technology will not achieve a feeling of satisfaction given the utilization challenges of such technology.
- B. Although most smart home technology is aimed at meeting or exceeding users' high expectations, those expectations in general remain poorly understood.
- C. Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.
- D. Although negative disconfirmation has often been studied, little is known about the cognitive mechanisms shaping users' reactions to it in the context of new technology adoption.

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues' conclusion?

- A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.
- B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.
- C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.
- D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

The ratio of methane to other atmospheric constituents—represented by a measure called the methane mole fraction—influences a variety of meteorological phenomena, notably precipitation and humidity. For Titan, Saturn’s largest moon, the observational data that exist are too sparse and discrepant to fully constrain the range of the methane mole fraction at various atmospheric levels. Juan Lora and colleagues point out that outputs of the IPSL atmospheric model of Titan, which track closely to observations in some respects, reflect how the model’s developers responded to this challenge: by prescribing a uniform methane mole fraction for the lowest level of the atmosphere. It is therefore important to note that _____ blank

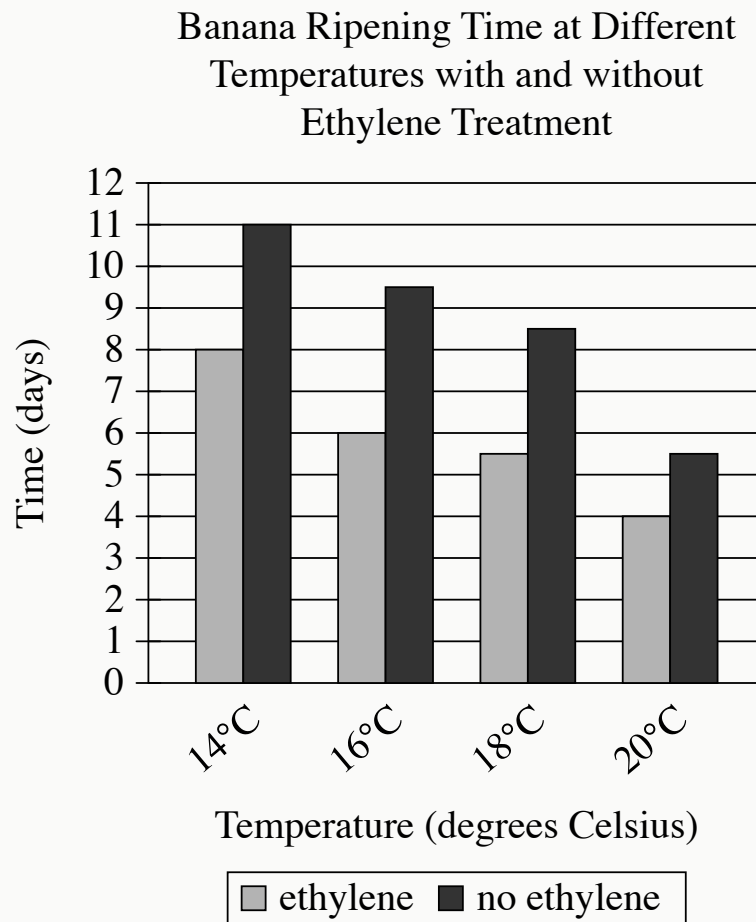
Which choice most logically completes the text?

- A. some disagreements between the model’s simulations of Titan’s precipitation and humidity and the moon’s actual precipitation and humidity are to be expected.
- B. further observations of Titan may clarify the moon’s methane mole fraction sufficiently for the model to employ a single value rather than a range.
- C. even though the model’s outputs sometimes agree with observational data, Titan’s real methane mole fraction is likely higher than the methane mole fraction used in the model.
- D. inconsistencies across the model’s simulations of Titan’s precipitation and humidity could be attributable to variations in the moon’s methane mole fraction.

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

- A. Dog breeds would not exist without many years of human intervention in dogs' reproduction.
- B. Research fails to confirm a commonly held belief about dog breeds and behavior.
- C. The dog breeds most popular among owners have often changed over the past 160 years.
- D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.



- For each data category, the following bars are shown:
 - ethylene
 - no ethylene
- All data are approximate.
- The Time data for the 4 categories are as follows:
 - 14 degrees Celsius:
 - ethylene: 8 days
 - no ethylene: 11 days
 - 16 degrees Celsius:
 - ethylene: 6 days
 - no ethylene: 9.5 days
 - 18 degrees Celsius:

- ethylene: 5.5 days
- no ethylene: 8.5 days
- 20 degrees Celsius:
 - ethylene: 4 days
 - no ethylene: 5.5 days

A student is conducting an experiment to test the effect of temperature and ethylene treatment on the ripening speed of bananas. The student treated some bananas with ethylene while leaving others untreated, then allowed both types of bananas to ripen at one of four different temperatures. Comparing the data for bananas with and without ethylene, the student concluded that _____blank

Which choice most effectively uses data from the graph to complete the student's conclusion?

- A. 20°C is the ideal temperature at which to store bananas to slow ripening time.
- B. for those bananas that were not treated with ethylene, differences in temperature were not associated with absolute differences in ripening time.
- C. bananas treated with ethylene ripen faster at 14°C and 16°C than at 18°C and 20°C.
- D. ethylene was associated with a greater absolute change in ripening time at 14°C, 16°C, and 18°C than at 20°C.

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because _____ blank

Which choice most logically completes the text?

- A. it can only be used on burrows below a certain size.
- B. it does not allow for multiple castings of the same burrow over time.
- C. the casting process takes more time than *A. bellulus* takes to construct a burrow.
- D. the process of recovering the model distorts the resin's shape.

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

- A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.
- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.
- C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.
- D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.